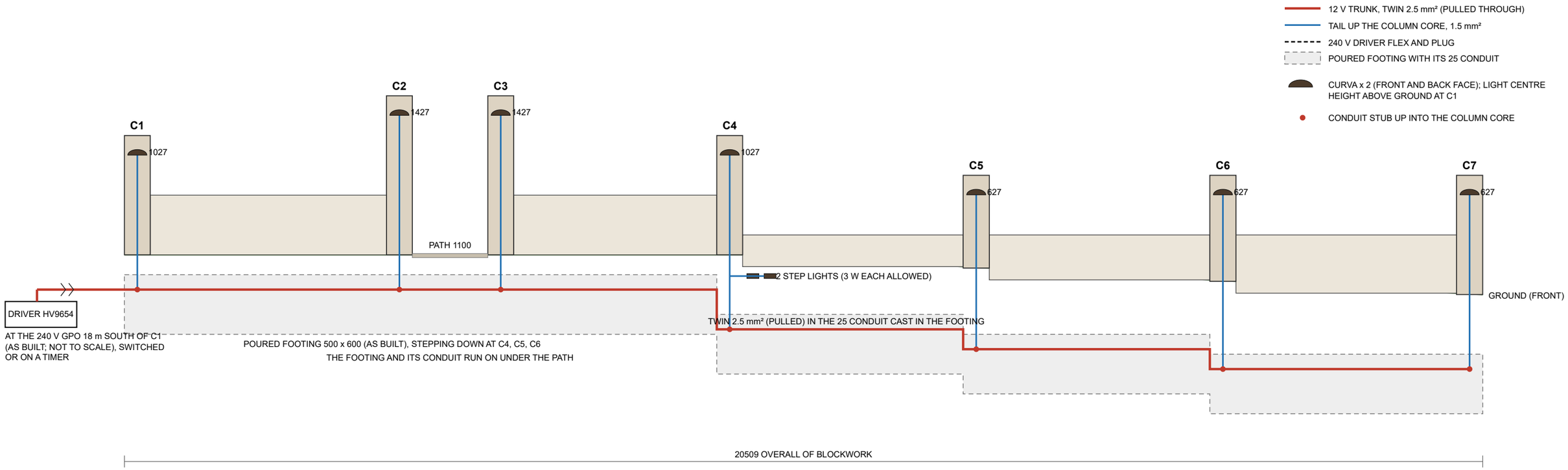
BLOCK WALL BW01 - COLUMN LIGHTS, HAVIT CURVA HV3251T-AB-12V: WIRING AND MOUNTING SCHEMATIC

PRELIMINARY - FOR THE ELECTRICIAN TO CONFIRM

13 NOV 2026 · set-out as built: opening 1140 between the C2 and C3 block faces; walls tiled 20 off the blocks; 25 conduit cast in the footings, twin 2.5 mm² pulled through, driver at the GPO 18 m south of C1, as built · schematic, not to scale except where stated

A. WIRING LAYOUT - WALL SEEN FROM THE FRONT (C1 LEFT). LENGTHS ALONG THE WALL TO 1:60, HEIGHTS DRAWN 1.5 TIMES AS TALL.



VOLTAGE AT THE LIGHTS ON THE 2.5 mm² AS PULLED (HAVIT: 11.40 V OR MORE, NO MORE THAN 25 m OF CABLE FROM THE DRIVER)

Column	Light centre (mm)	As installed: driver 18 m south, step lights on (volts / cable)	Same, step lights fed separately	If 240 V is brought to the wall at C4 (volts / cable)
C1	1027	**10.92 V / 21 m**	**11.06 V**	11.78 V / 12 m
C2	1427	**10.73 V / 25 m**	**10.88 V**	11.80 V / 9 m
C3	1427	**10.66 V / 27 m**	**10.83 V**	11.82 V / 7 m
C4	1027	**10.53 V / 30 m**	**10.72 V**	11.90 V / 3 m
C5	627	**10.45 V / 34 m**	**10.64 V**	11.82 V / 7 m
C6	627	**10.39 V / 38 m**	**10.59 V**	11.77 V / 11 m
C7	627	**10.36 V / 42 m**	**10.56 V**	11.74 V / 15 m

AS INSTALLED THIS FAILS HAVIT'S RULES: every light is below 11.40 V and C2 to C7 are more than 25 m of cable from the driver; C7 gets about 10.4 V at 42 m. For the electrician to resolve before the lights are fitted. A bigger driver does not help (it still puts out 12 V); bringing 240 V to the wall does.

THE WORKING (AS INSTALLED, STEP LIGHTS ON)

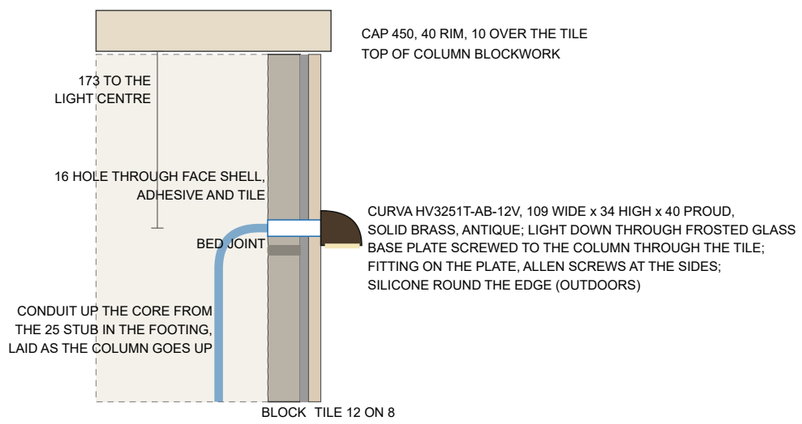
Trunk run	Length	Current
Driver (18 m south) to the C1 stub	18.85 m	4.00 A
C1 to C2	3.96 m	3.50 A
C2 to C3, on under the path	1.53 m	3.00 A
C3 to C4	3.86 m	2.50 A
C4 to C5	3.92 m	1.50 A
C5 to C6	3.92 m	1.00 A
C6 to C7	3.73 m	0.50 A

Each Curva 3 W = 0.25 A at 12 V; a tail feeds 2 lights. Drop out and back = 2 × 0.0175 × length × current ÷ 2.5 mm², added run by run; tails 1.5 mm². Lengths follow the footing's steps; the conduit's real depth may add a little. The driver's size does not change these figures: a 12 V constant voltage driver puts out 12 V whatever its rating; the drop comes from the current and the cable.

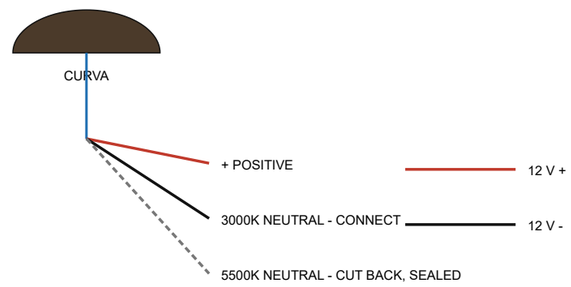
NOTES

1. 14 × Havit Curva HV3251T-AB-12V (solid brass, antique brass), 3 W, 240 lm at 3000K, IP65, 12 V DC, on both faces of each column, bottom of each light on the bed joint under the top block (light centre 173 below the top of the column blockwork); see the drill pattern, sheet BW01-5. Wired fixed at warm white 3000K.
2. Driver: 1 × Havit HV9654, 100 W weatherproof (IP66), 12 V DC constant voltage, flex and plug, at the 240 V outdoor GPO 18 m south of C1 (as built). Load: 14 × 3 W + 2 step lights allowed at 3 W = 48 W; Havit sizes drivers for light fittings at 50% of their rating, so 100 W (their recommended maximum load 80 W); a 150 W driver also suits but does not raise the voltage at the lights. Switched or on a timer on the 240 V side; not dimmed.
3. As built: 25 conduit cast in the footings, up into every column core (C1 to C7) and out of the footing at the power point, with twin 2.5 mm² pulled through, west to C1 and east to C7. The footing runs on under the path, so the cable crosses the C2 to C3 opening inside it.
4. From each stub, carry the conduit up the column core to both light positions as the column goes up, before core fill, clear of the N12 bars - certifier to accept. 1.5 mm² tails in it to the lights.
5. Make every joint where it can be reached again (behind a fitting or in a draw box), never buried in core fill or the footing. Havit joints: solder, then adhesive-lined heat shrink or a gel cap, or Havit IP67 connectors.
6. Havit: installed by a licensed electrician to AS/NZS 3000; mains off before work; gloves when handling brass fittings; silicone round each fitting outdoors. Havit lists aluminium, 304 stainless, powder-coated brass and 316 stainless fittings as not for use within 5 km of salt water, pools included; the antique brass Curva is solid brass - confirm with Havit for this site.
7. The 2 step lights on the step riser by C4: fitting not chosen; on this circuit they cost about 0.2 V at the far lights.

B. MOUNTING DETAIL - SECTION THROUGH A COLUMN FACE AT A LIGHT, 1:6



C. WIRING AT EACH LIGHT (HAVIT), FIXED AT WARM WHITE 3000K



ALL 16 LIGHTS IN PARALLEL ON THE ONE 12 V DC DRIVER. NEVER CONNECT A LIGHT TO 240 V.
JOINTS SOLDERED, COVERED WITH ADHESIVE-LINED HEAT SHRINK OR A GEL CAP,
OR HAVIT IP67 CONNECTORS HV9911 / HV9912 / HV9913 (HAVIT).
4000K WOULD JOIN BOTH NEUTRALS; NOT USED HERE.